

Autonomous Systems

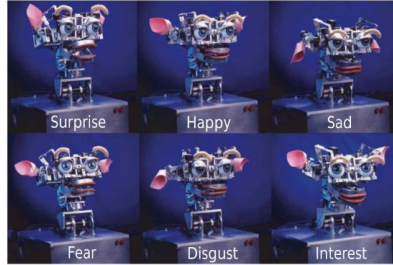
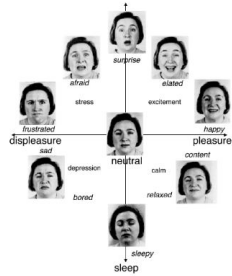
Week 09: Cognitive (developmental) Robotics

Murat Kirtay

<http://www.crossvalidate.me/>

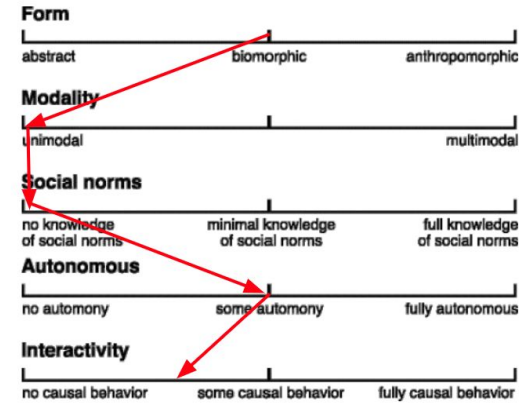
Previously in Autonomous Systems

Emotions: from humans to robots



“emotion-inspired mechanisms can **improve** the way autonomous robots operate in a human environment with people, and can improve the ability of these robots to effectively achieve their own goals.”

Social robotics

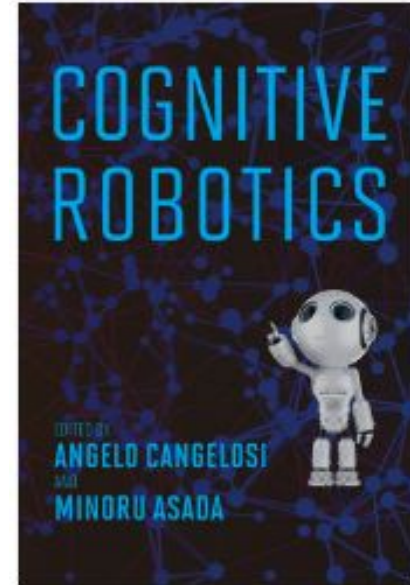


Questions from previous lecture



Weeks	Lecture topics
01	Intro to Autonomous Systems & Course Logistics
02	Robotics: Sensors and Actuators. & Braitenberg Vehicles
03	Controllers (Brains) in Autonomous Systems
04	Embodiment in Autonomous systems
05	Adaptivity and Reinforcement Learning in Autonomous Systems
06	Unsupervised Learning in Autonomous Systems
07	Multimodal Learning, Interaction, and Communication
08	Affective and social Human-Robot and Robot-Robot Interactions
09	Cognitive Robotics
10	Simulation environments for autonomous systems
11	Selected topics in autonomous systems
12	Invited Speaker I
13	Invited Speaker II [cancelled]

Today's Agenda



Vernon's Cognitive Robotics Course

Cognitive Robotics



An introductory course for the IEEE Robotics and Automation Society Technical Committee for Cognitive Robotics



Instructors: send an [email](#) to request the complete set of PowerPoint slides. Example code is available on [Github](#).
If you already have a copy, check the [Version History](#) to make sure you have the most recent version. If you don't, please send an email to request it.



A PR2 robot pours popcorn from a saucepan (left) and sets a table (right) during demonstrations of cognitively-enabled robot manipulation using CRAM.
Image courtesy of the Everyday Activity Science and Engineering (EASE) interdisciplinary research center at the University of Bremen, Germany.

Content of this presentation *partially* adapted from David Vernon's lectures on Cognitive Robotics.

Mandatory Quote



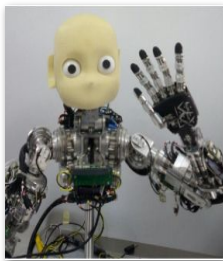
Instead of trying to produce a programme to **simulate the adult mind**, why not rather try to produce one which simulates the **child's**? If this were then subjected to an appropriate course of education, one would obtain the adult brain [...]
Our hope is that there is so little mechanism in the child brain that something like it can be easily programmed. The amount of work in the education we can assume, as a first approximation, to be much the same as for the human child.

Mandatory Quote



Instead of trying to produce a programme to **simulate the adult mind**, why not rather try to produce one which simulates the **child's**? If this were then subjected to an appropriate course of education, one would obtain the adult brain [...] **Our hope is that there is so little mechanism in the child brain that something like it can be easily programmed.** The amount of work in the education we can assume, as a first approximation, to be much the same as for the human child.

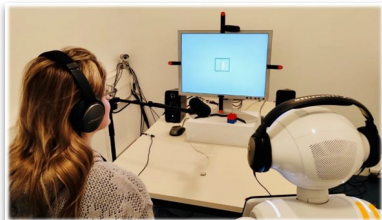
Inspiration for **CoDeRs: Cognitive Developmental Robotics**



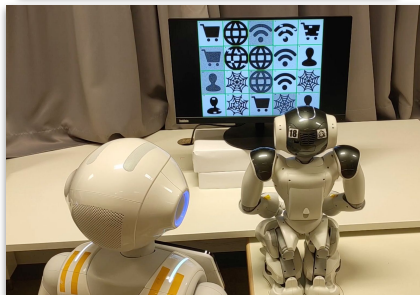
**ROBOTS
WITHOUT COGNITION**



**COGNITION
WITHOUT ROBOTS**



**COGNITIVE
ROBOTICS**



**COGNITION-ENABLED
ROBOT-ROBOT
INTERACTION**

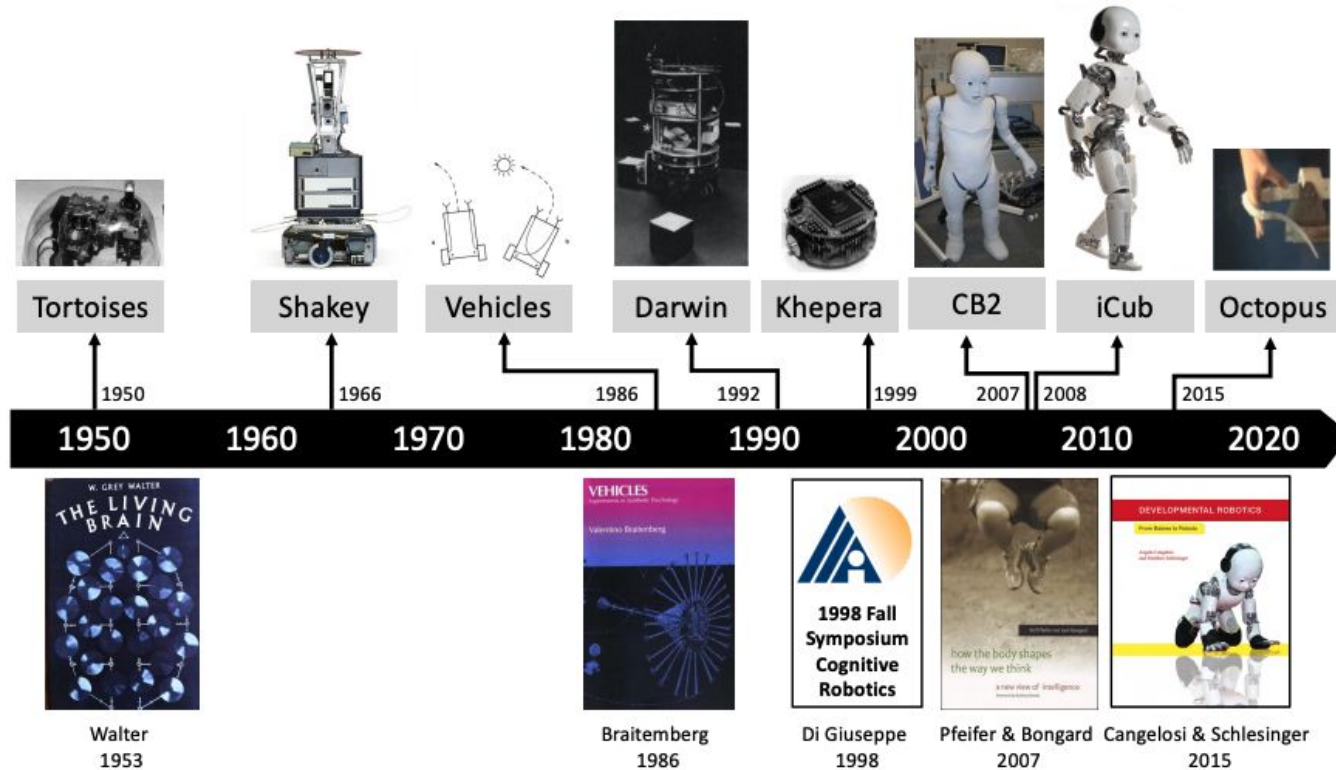


What is cognitive system (or cognition)?

Researchers' definitions

- Cognition is the ability to relate **perception** and **action** in a **meaningful** way determined by experience, learning and memory. *Mike Denham*
- A cognitive system possesses the ability of **self-reflection** (or at least self-awareness). *Horst Bischof*
- Cognition is **gaining knowledge** through the **senses**. *Majid Mermehdi*
- Cognition is the ability to **ground perceptions** in concepts together with the ability to **manipulate** concepts in order to **proceed** toward **goals**. *Christian Bauckhage*
- An artificial cognitive system is a system that is able to **perceive** its **surrounding environment** with multiple **sensors**, merge this information, **reason** about it, learn from it and **interact** with the outside world. *Barbara Caputo*
- Cognition is **self-aware** processing of information. *Cecilio Angulo*
- Cognitive Systems are ones that are able to extract and (most importantly) represent useful aspects of largely redundant, possibly irrelevant **sensory information** in a form that is most conducive to achieving a particular **high level goal**. *Sethu Vijayakumar*
- A cognitive system is a system that can **change its behaviour** based on **reasoning**, using **observed evidence** and domain knowledge. *Bob Fisher*
- Cognition is when I **know** what I **am doing**, when I can **judge** how good or bad it is, and **explain** why I am doing it. *Markus Vincze*
- Cognition is the ability to **plan**, **reason**, adapt and **act** according to high level **motivations or goals** and using a range of **senses**, typically including **vision**, and may be communicate. *Patrick Courtney*
- A cognitive system is an **autonomous anti-entropy** engine. *David Vernon*

Cognitive robotics timeline



What is cognitive system?

- A cognitive system is an **autonomous anti-entropy** engine. *David Vernon*



In-class activity: To cognize, or not to cognize

Which of the following robots is an example of cognitive robot? And explain why?



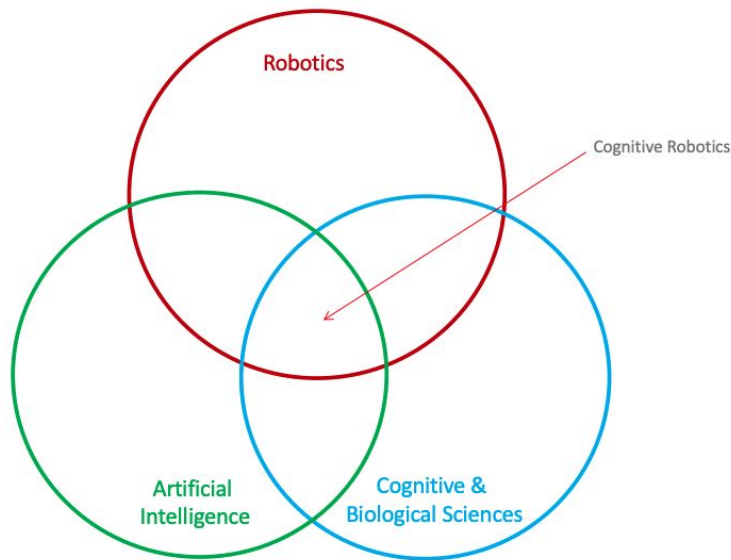
What is cognitive system?

- A cognitive system is an **autonomous anti-entropy** engine. *David Vernon*



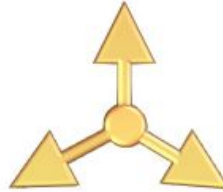
The definition that we will follow for cognitive robotics

*“Cognitive robotics is the field that combines insights and methods from **AI**, as well as **cognitive** and **biological** sciences, to robotics.”*



Cognitive robotics

Bio-inspired
human-like and animal-like
behaviour and intelligence

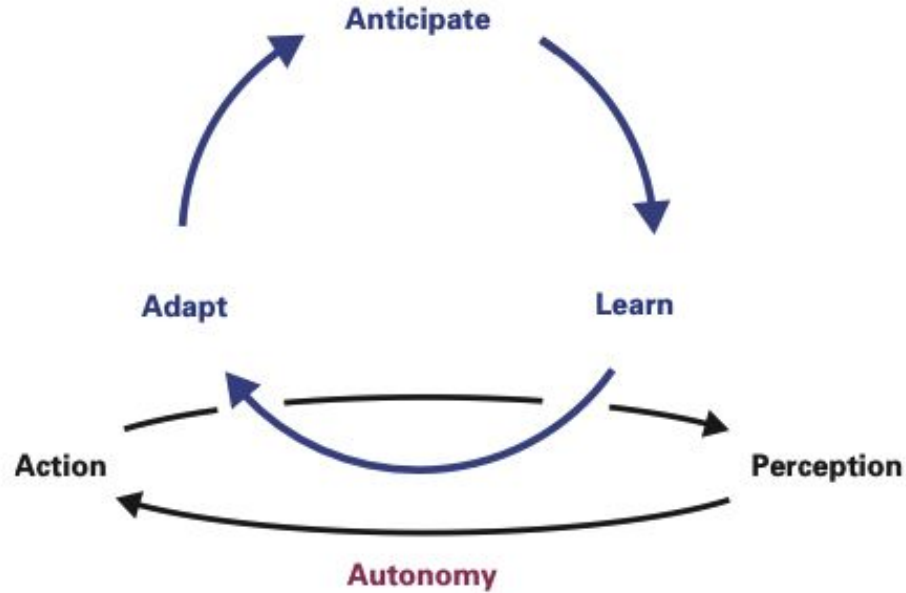


System-level integration
of a range of cognitive abilities:

- Sensorimotor skills
- Knowledge representation & reasoning
- Social interaction

Interdisciplinary approach,
including cognitive (neuro)science,
cognitive psychology, and biology

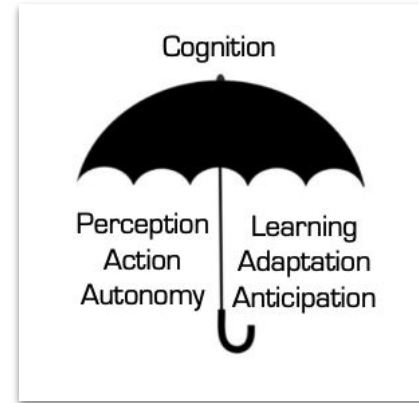
The six attributes of cognition



Two views of cognition



The Thinker
Auguste Rodin



Classical AI approach:
intelligence as computation (cognitivism)

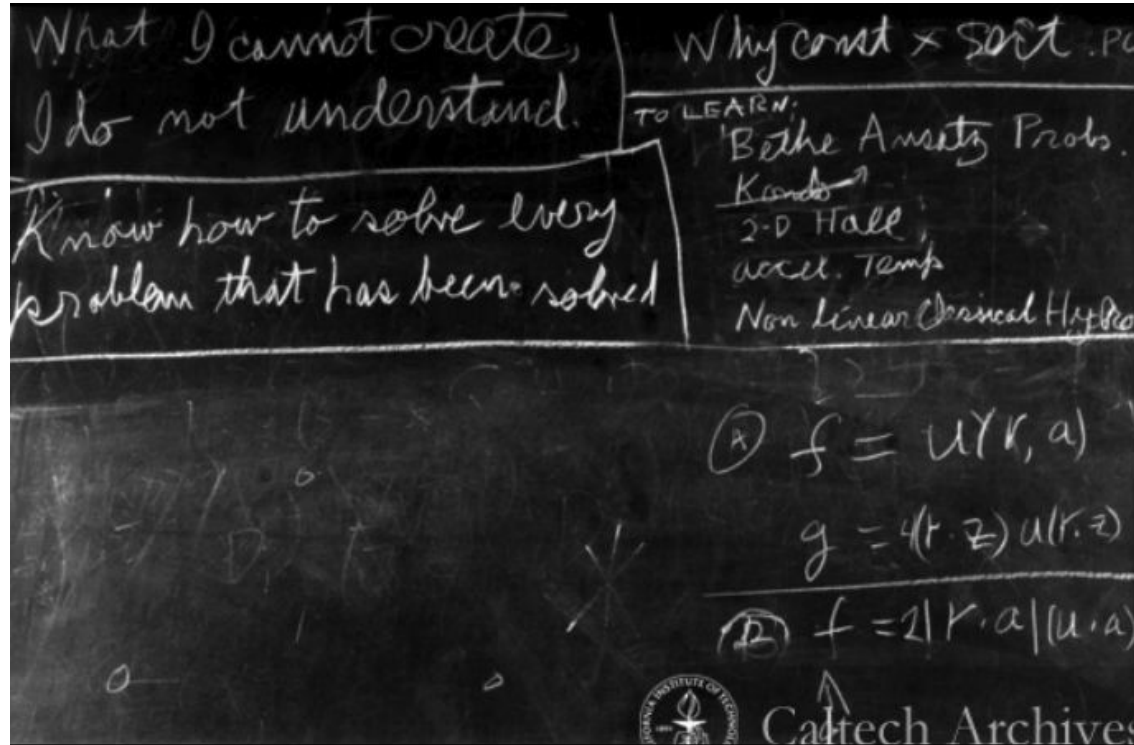


Embodied AI approach:
intelligent behaviours emerge from the
bodily interaction with the environment



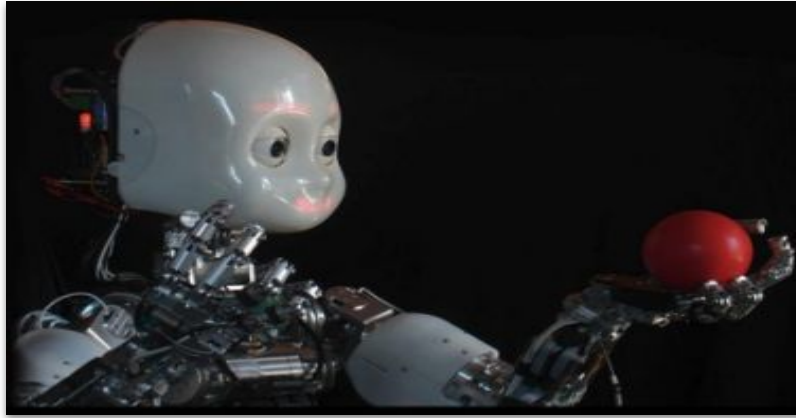
Pfeifer, R., & Bongard, J. (2006). *How the body shapes the way we think: a new view of intelligence*. MIT press.

How can I create/built a cognitive robot.



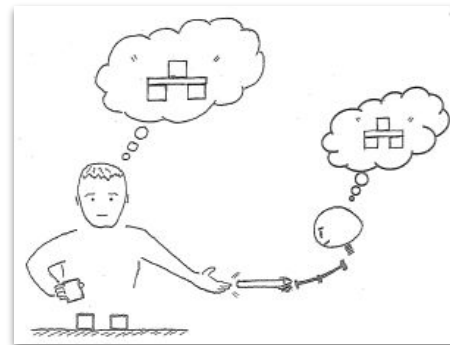
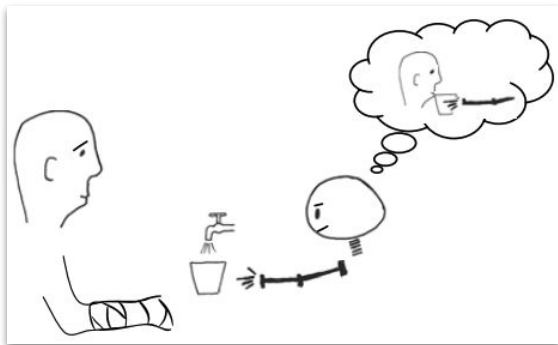
Remember synthetic methodology

Cognitive robots achieve their goals by



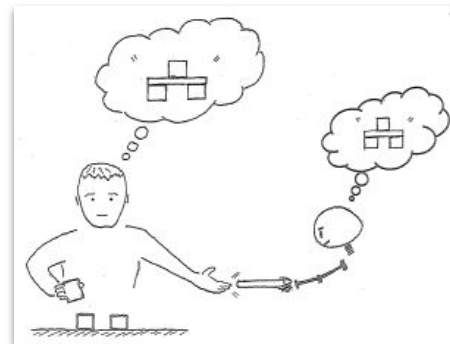
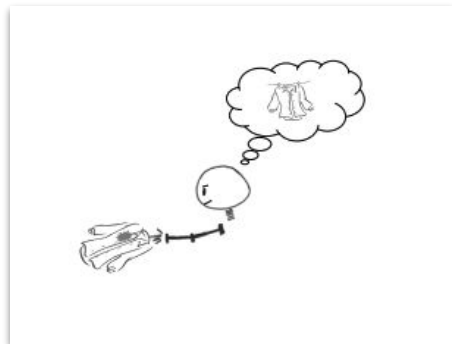
- **Perceiving** their environment
- Paying **attention** to the events that matter
- **Anticipating** the need for some action
- **Planning** what to do
- **Anticipating** the outcome as it executes the action
- **Learning** from the resultant interaction
- **Adapting** to change

Why is cognition useful for robots?



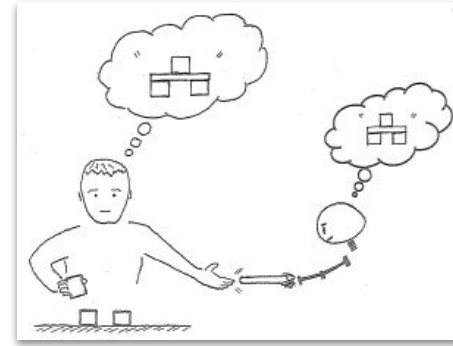
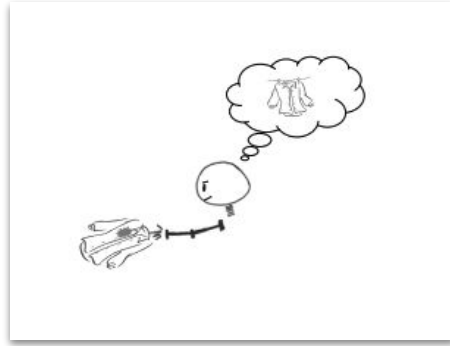
1. It allows robots to work **autonomously** in everyday environments
 - a. **anticipating** outcomes when selecting the actions it will perform
 - b. **adapting** to changes and unforeseen situations
2. It enables effective **interaction** with humans
 - a. A cognitive ability can enable a robot to **infer the goals and intentions** of the person with whom it is interacting and thereby allows it to behave in a helpful manner.
 - b. Humans like to **engage with other cognitive agents**: if a robot has a capacity for cognition it fosters human robot interaction

Why is cognition useful in robotics?



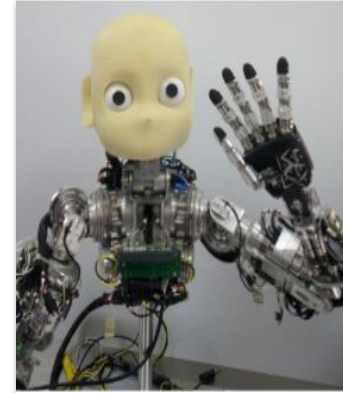
- Core cognitive abilities
 - perception,
 - attention,
 - action selection,
 - memory,
 - learning,
 - reasoning,
 - Metacognition (thinking about thinking),
 - Prospection (evaluation of future)

Why is cognition useful in robotics?



- Core cognitive abilities

- perception,
- attention,
- action selection,
- memory,
- learning,
- reasoning,
- Metacognition (thinking about thinking),
- Prospection (evaluation of future)



Is it possible to integrate all these abilities into a single architecture?

Desirable Characteristics

Desideratum 1. Value systems and motives

Desideratum 2. Physical embodiment

Desideratum 3. Sensorimotor contingencies

Desideratum 4. Perception

Desideratum 5. Attention

Desideratum 6. Prospective action

Desideratum 7. Declarative and procedural memory

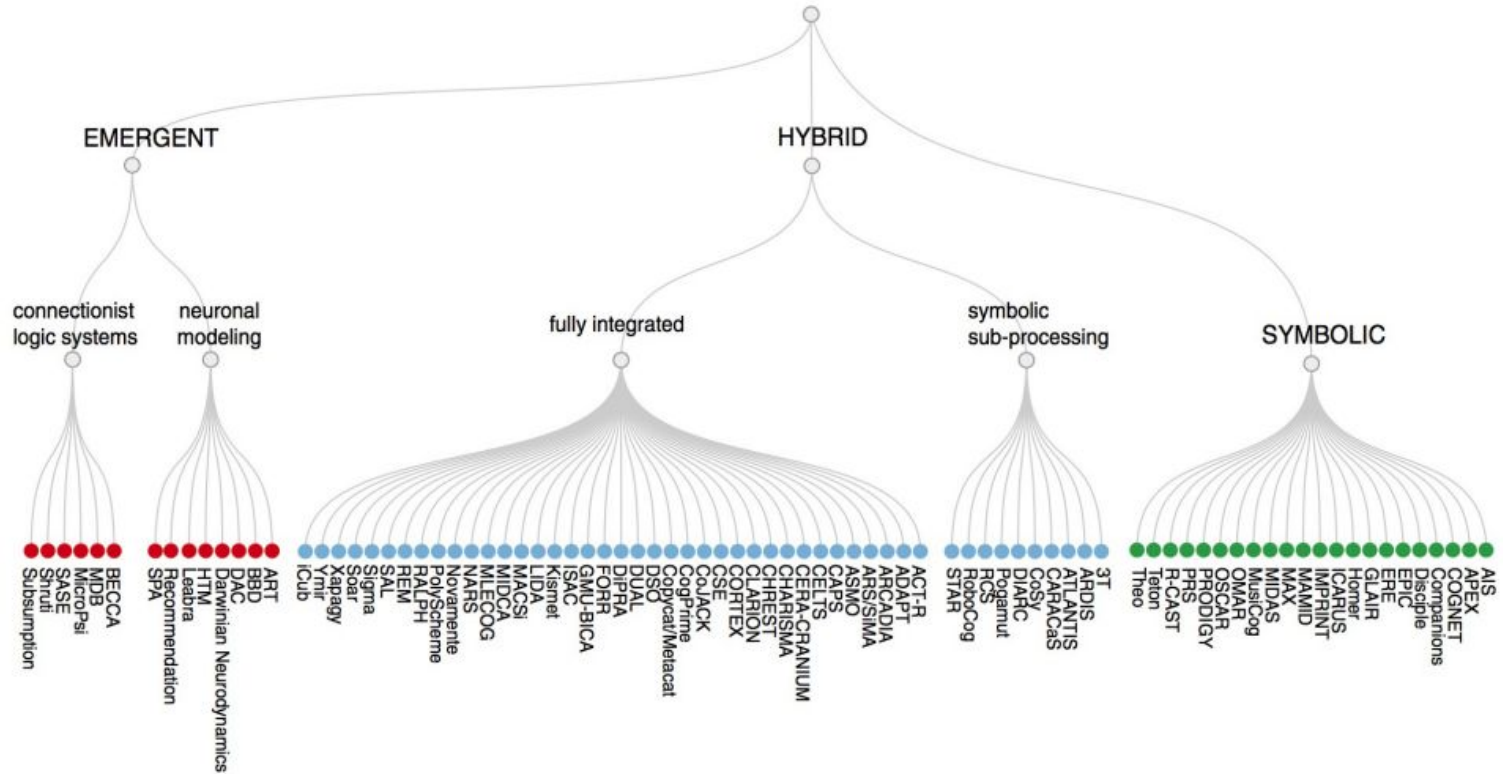
Desideratum 8. Multiple modes of learning

Desideratum 9. Internal simulation

Desideratum 10. Constitutive autonomy

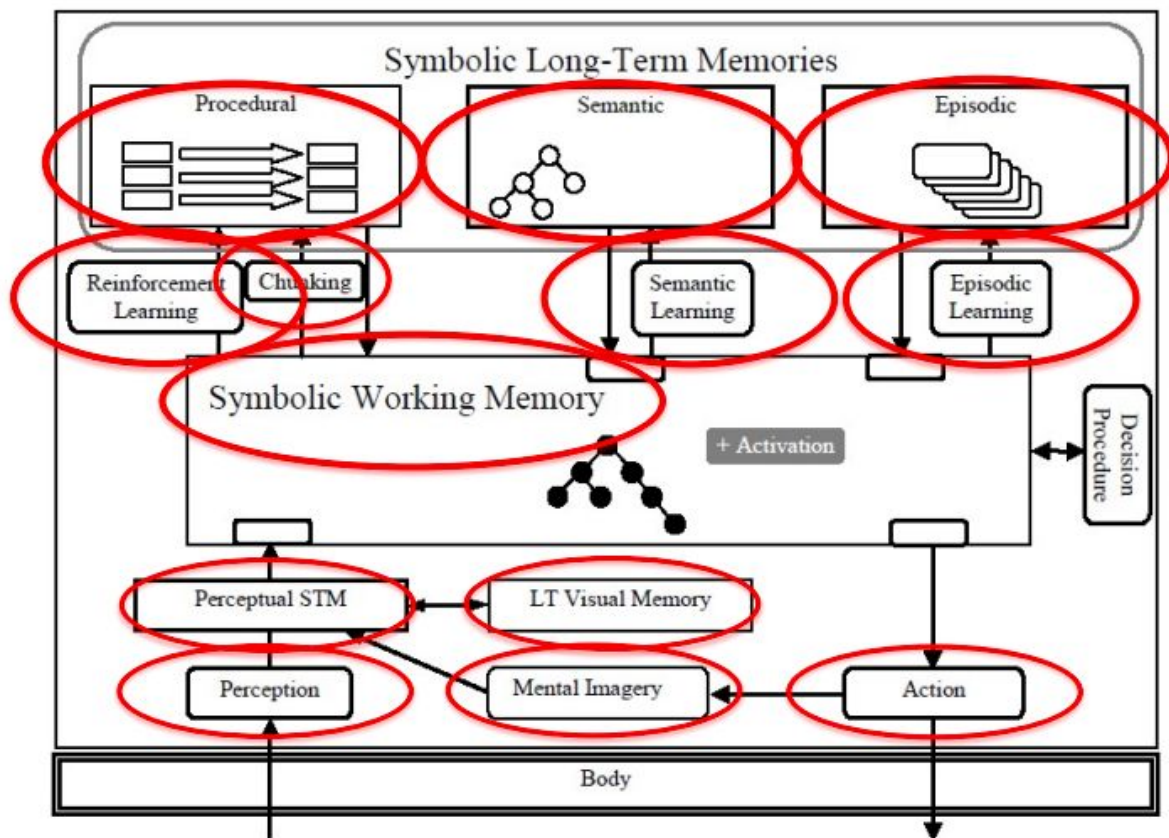


Taxonomy of cognitive architectures



Soar

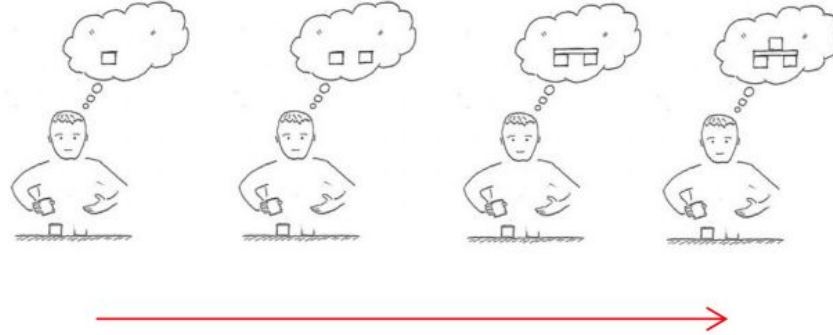
[Laird et al. 2012]



Cognitive systems continually **predict**

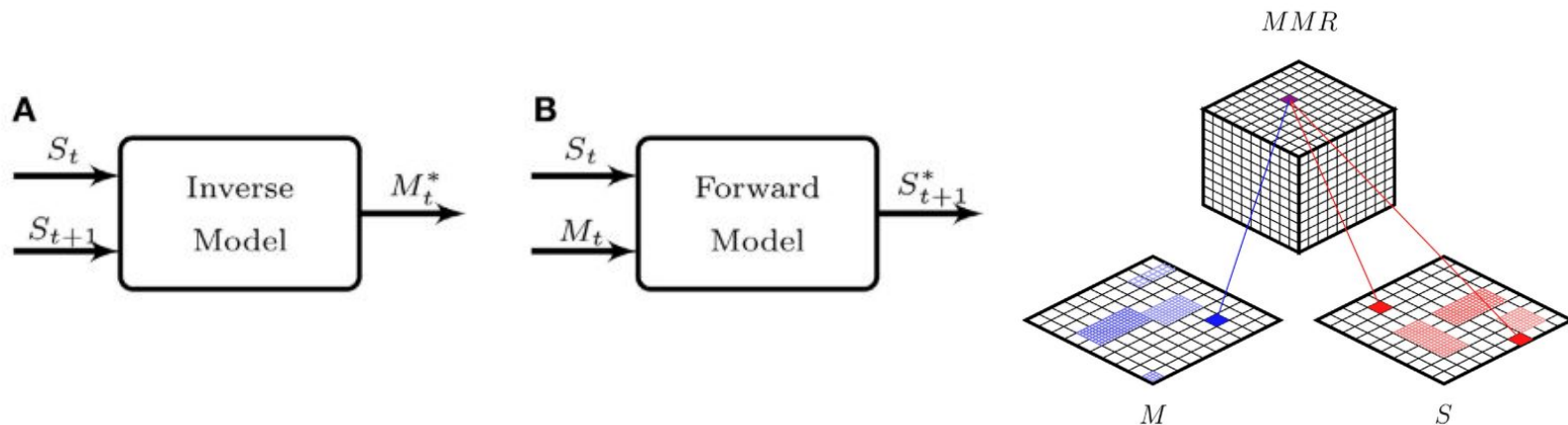
The need for **action**
(self and others)

The **outcome** of those actions



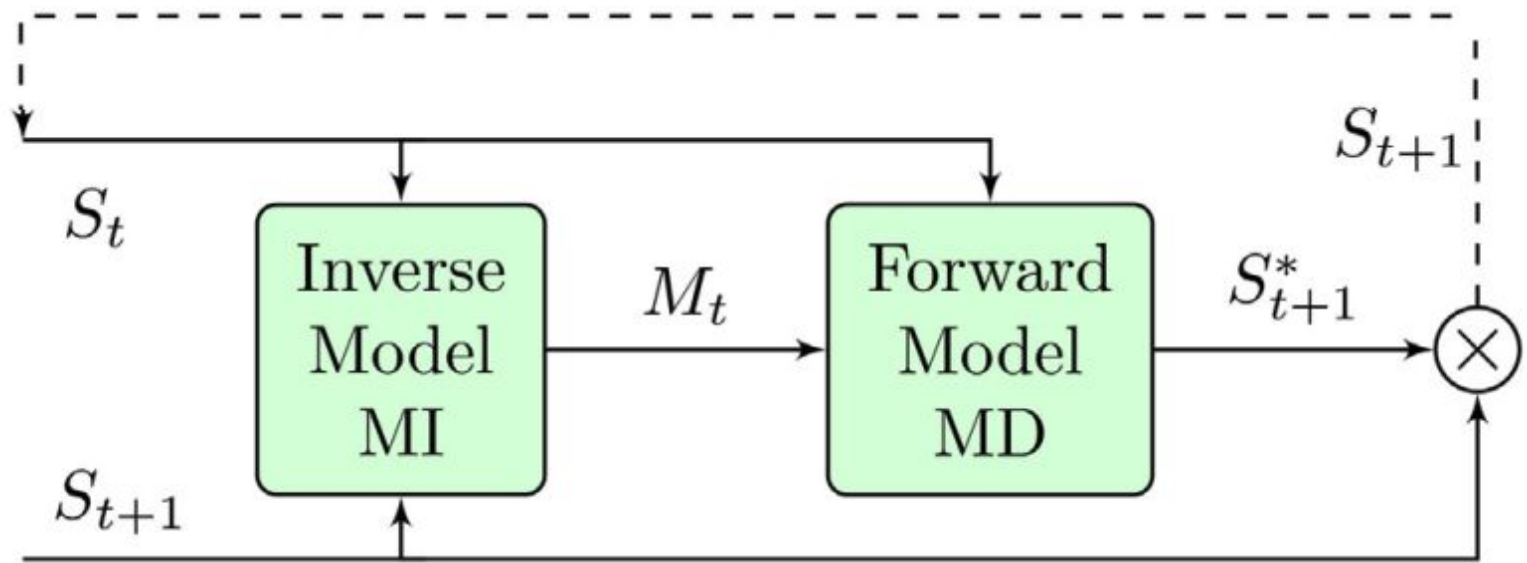
Ok! But how!

Self-Organized Internal Models Architecture

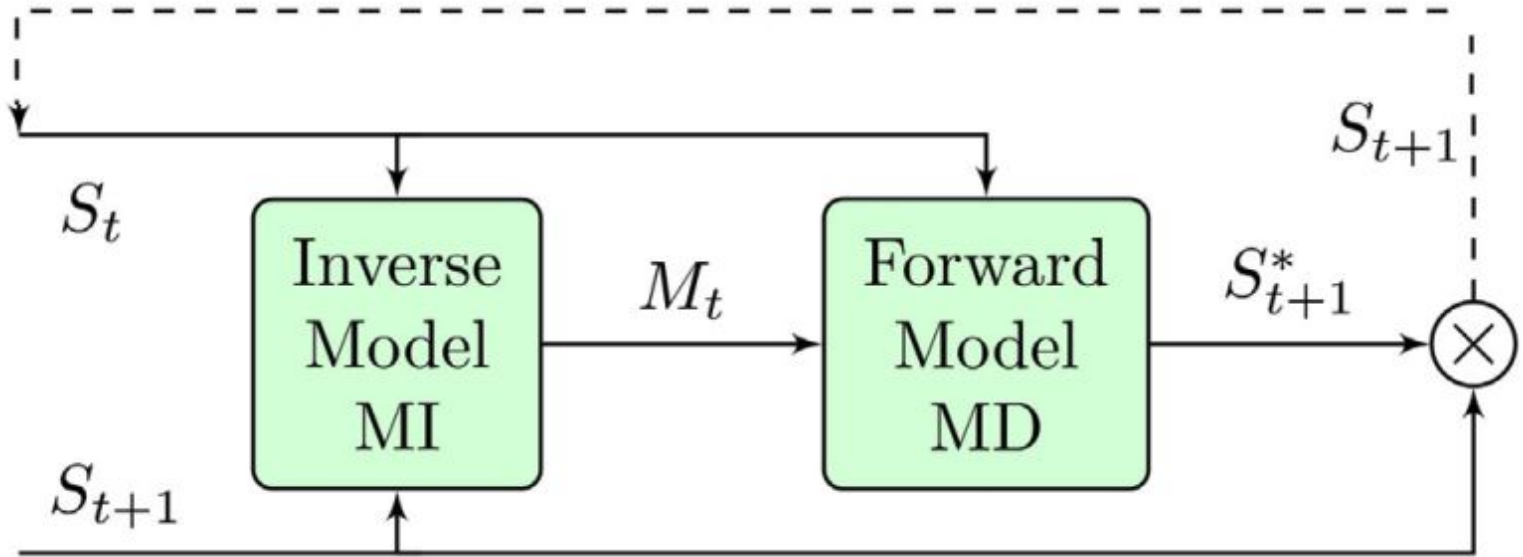


SOIMA Architecture

- The units of the architecture are self organizing maps
- Implementation of inverse and forward models in the same architecture
- Missing modality representation can be constructed
- Sensory motor mapping



Inverse model: given the current sensory state S_t and a desired sensory state for time $t + 1$, S_{t+1} this model suggests the motor command M_t^* that brings about the change in the sensory state of the agent. (B) Forward model: given the current sensory state S_t and a motor command M_t , the forward model predicts the resulting *sensory state prediction* S_{t+1}^* , were the command executed. It is worth noting the difference between S_{t+1}^* and S_{t+1} . The former is the actual expected state for the next time-step, but the latter may require several time-steps in order to be attained.



You can also employ these concepts for your project.

Why we study cognitive robotics?



There are **two** reasons people study cognitive robotics

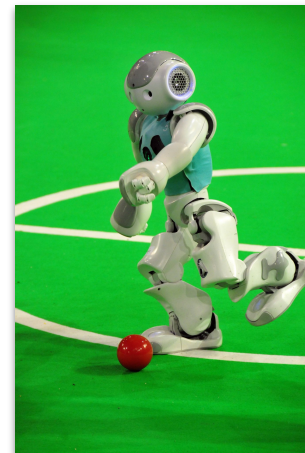
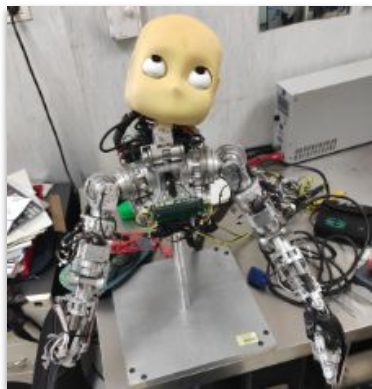
1. They want to build smart robots
2. They want to understand cognition

Why we study cognitive robotics?



There are **two** reasons people study cognitive robotics

1. They want to build smart robots
 2. They want to understand cognition
- + **Cognitive robotics is Fun!**



Cognitive robotics beyond research labs

Industrial Priorities for Cognitive Robotics

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Rwanda
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Markus Vincze
Technische Universität Wien
Austria
Email: vincze@acin.tuwien.ac.at

Abstract—We present the results of a survey of industrial developers to determine what they and their customers require from a cognitive robot. These are cast as a series of eleven functional abilities:

- 1) Safe, reliable, transparent operation.
- 2) High-level instruction and context-aware task execution.
- 3) Knowledge acquisition and generalization.
- 4) Adaptive planning.
- 5) Personalized interaction.
- 6) Self-assessment.
- 7) Learning from demonstration.
- 8) Evaluating the safety of actions.
- 9) Development and self-optimization.
- 10) Knowledge transfer.
- 11) Communicating intentions and collaborative action.

three parameters: safety for people, safety for equipment, and safety of the robot system. The rules may change depending on the environment and a cognitive robot will not exceed the limits of safe operation. The limits may be application specific, e.g., the robot should not deviate further than a given specification/distance/etc. A cognitive robot will use this type of knowledge to act responsibly and will ask for assistance when necessary (e.g. before it encounters difficulties). In particular, in emergency situations, the robot will stop all tasks to follow some emergency procedure. Ideally, if the user is deliberately trying to misuse the robot, e.g. programming it to assist with some unethical task, a cognitive robot will cease operation.

More on Cognitive Architectures

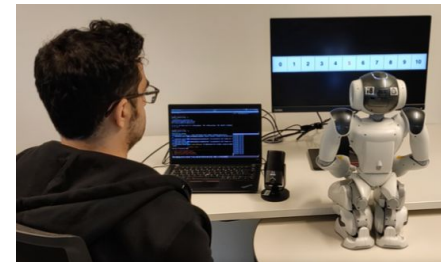
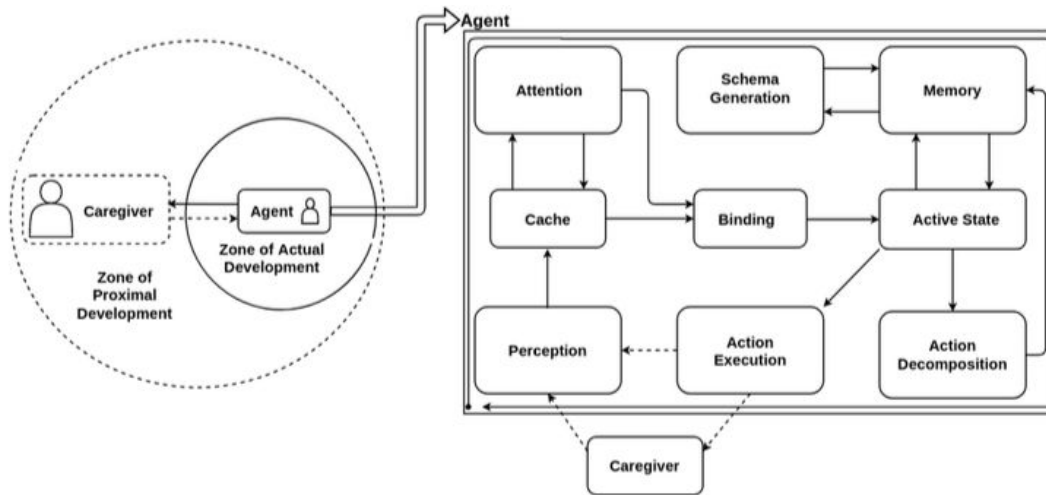
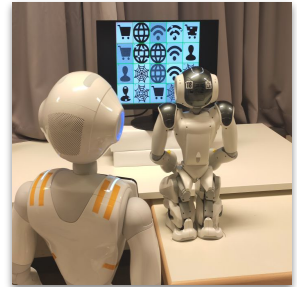
Learning in Social Interaction: Perspectives from Psychology and Robotics



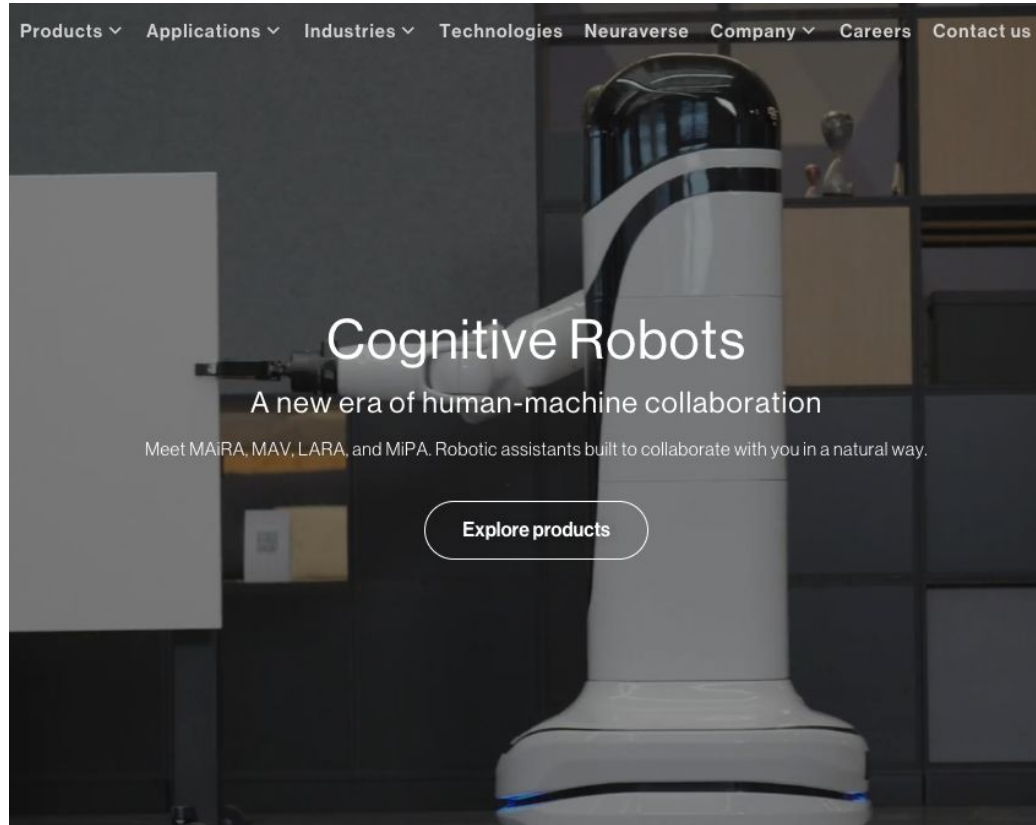
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Cognitive robotics beyond research labs



Developmental trajectory of human infants

PHYSICAL DEVELOPMENT	Average age skills begin	3 months	6 months	9 months	1 year	2 years	3 years	5 years
Head and trunk control	 lifts head part way up	 holds head up briefly	 holds head up high and well	 holds up head and shoulders	 turns head and shifts weight	 holds head up well when lifted	 moves and holds head easily in all directions	
Rolling		 rolls belly to back	 rolls back to belly	 rolls over and over easily in play				
Sitting		 sits only with full support	 sits with some support	 sits with hand support	 begins to sit without support	 sits well without support	 twists and moves easily while sitting	
Crawling and walking		 begins to creep	 scoots or crawls	 pulls to standing	 takes steps	 walks	 runs	 can walk on tiptoe and on heels
Arm and hand control	 grips finger put into hand	 begins to reach towards objects	 reaches and grasps with whole hand	 passes object from one hand to other	 grasps with thumb and forefinger	 easily moves fingers back and forth from nose to moving object	 throws and catches ball	
Seeing	 follows close object with eyes	 enjoys bright colors/shapes	 recognizes different faces	 eyes focus on far object	 looks at small things/pictures	 Sees small shapes clearly at 6 meters (see p. 453 for test).	 Sees small shapes clearly at 6 meters (see p. 453 for test).	
Hearing	 moves or cries at a loud noise	 turns head to sounds	 responds to mother's voice	 enjoys rhythmic music	 understands simple words	 hears clearly and understands most simple language		

Developmental robotics

Infant developmental behavior and learning targets

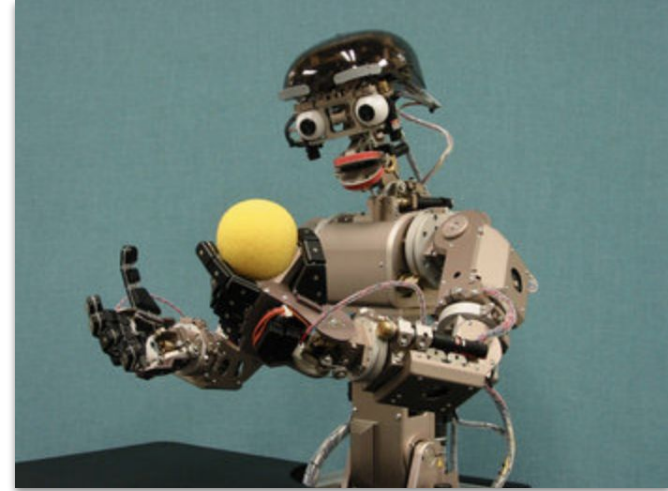
Month—behavior	Learning targets
5 hand regard	Forward and inverse models of the hand
6 finger the another's face	Integration of visuotactile sensation of the face
7 drop objects and observe the result	Causality and permanency of objects
8 hit objects	Dynamic modeling of objects
9 drum or bring a cup to mouth	Tool use
10 imitate movements	Imitation of unseen movements
10 rudimentary sympathy	Feel pain and empathy
11 grasp and carry objects to others	Action recognition and generation, cooperation
12 pretend	Mental simulation

The learning targets are quite similar to the tasks we would like to achieve with robots, autonomously.

Characteristics of human development

“Human development is characterized by **online**, **crossmodal**, **continuous**, **open-ended** learning.

- “**Online**” refers to the fact that learning happens while the child interacts with the environment and not in an offline mode.
- “**Crossmodal**” refers to the fact that different modalities and cognitive domains are acquired in parallel by the child and interact with each other.
- “**Continuous**” and “**open-ended**” refers to the fact that learning and development do not start and stop at specific stages but rather are lifelong learning experiences,”



What can we learn from children?



What can we learn from chimps?



How about cognitive robots?

The image is a composite graphic with a black background. At the top center, the text "Altruistic Behavior by 14-month-old Infants" is displayed in white, with "W. Wameken & M. Tomasello, 2008" in smaller text below it. The main visual is a video frame showing a 14-month-old infant in a blue jacket sitting at a table with a purple cloth, looking at a stack of blue blocks. An adult's arm is visible behind the child. In the top right corner of the video frame, there is a small inset showing a person's profile. In the bottom right corner of the video frame, the text "© Wameken & Tomasello" is visible. On the left side, there is a small inset image of a woman with dark hair, wearing a dark jacket, with the "EEB" logo above her. In the top left corner, there is a circular logo with the letters "HP" inside, with the text "Human Brain Project Education Programme" below it. At the bottom, there are four logos: the "EEB" logo, the "Human Brain Project" logo, the "EBRAINS" logo, and the European Union flag.

HP
Human Brain Project
Education Programme

EEB

Altruistic Behavior by 14-month-old Infants
W. Wameken & M. Tomasello, 2008

© Wameken & Tomasello

EEB

Human Brain Project

EBRAINS

European Union

Cognitive Development

Development arises due to changes in the central nervous system as a result of dynamic interaction with the environment

- Emergence of new forms of action
- Acquisition of predictive control of these actions

Mastery of action relies critically on prospection

- Inference of upcoming events
- Inference of outcome of actions

Can't we just learn?

Why we need development?

Learning vs. Development

Learning

A process for improving the performance of a system

A process for estimating or improving the parameter values that govern the behaviour of a **known model**

Development

A process for **generating** or discovering the **model** itself

Requires two-way interaction between agent and world

Requires **scaffolding on existing capabilities**

Remark: unsupervised learning and reinforcement learning do this to an extent



Cognitive Development

Discover new ways of doing things

- Typically make use of **scaffolding**
- The progressive development of innate skills or skills that are learned early on
- Development is phased

Developing the ability to help others and be helped by others

- 14 to 18 months: **instrumental helping behaviour**
- 2 years: start to solve **simple cooperation** tasks with adults
- 2-3 years: ability to **cooperate** with peers
- 3 years: complex **collaboration**
 - Sharing of intentions
 - Joint coordination of complementary actions
- 3 ½ years: Roles in the task can be reversed; & can teach new partners

Suggested videos on cognitive robotics



<https://transair-bridge.org/workshop-2021/>